

New Horizons for High Redshifts
IoA Cambridge, 27.07.11

Probing reionisation with the Ly- α Forest



Jamie Bolton

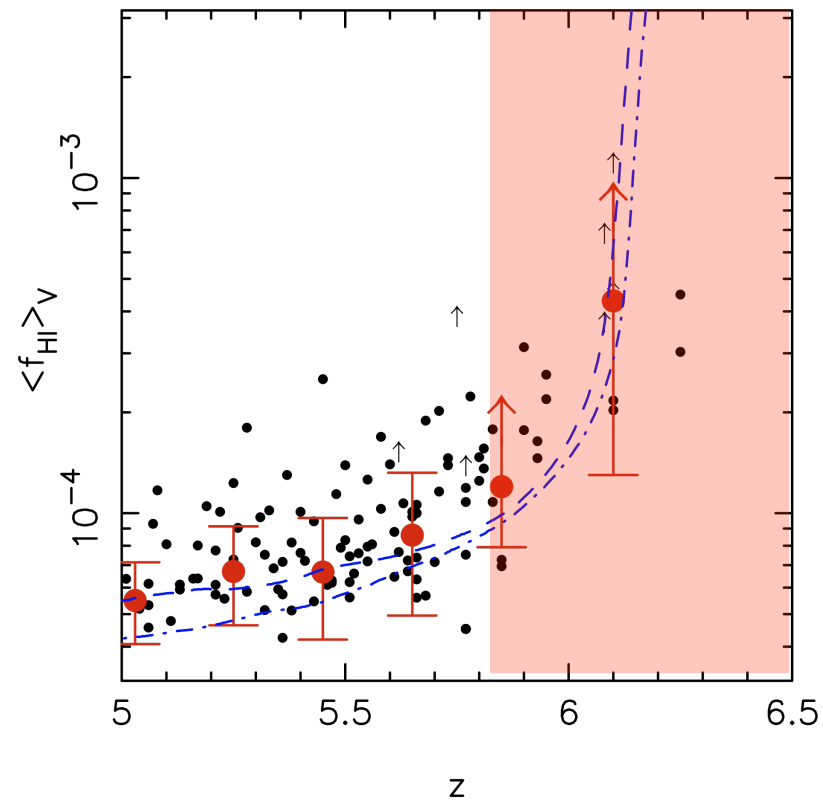
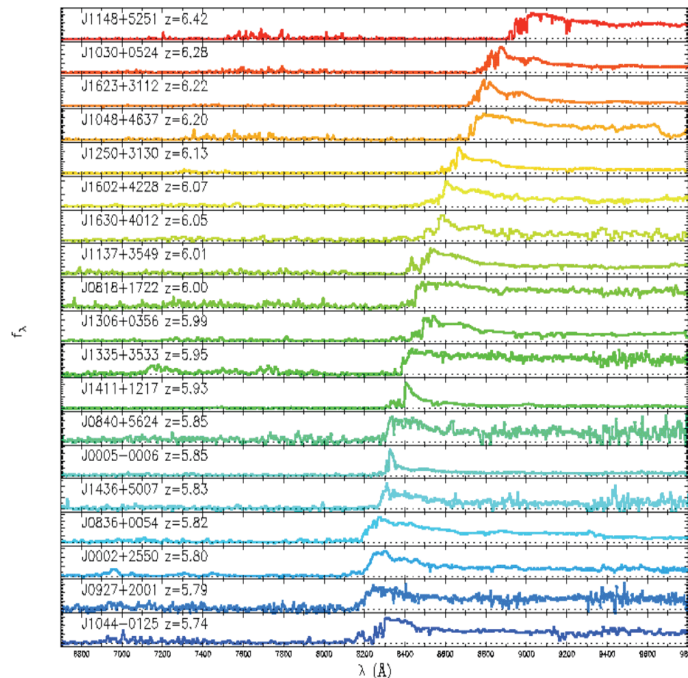
with thanks to Stuart Wyithe (Melbourne),
George Becker, Martin Haehnelt (IoA/KICC, Cambridge)
Steve Warren, Daniel Mortlock (Imperial) & UKIDSS et al.

Evidence from the Ly- α Forest

I: The Gunn-Peterson trough

The Gunn-Peterson (1965) trough at $z > 6$ implies f_{HI} is increasing towards higher redshift, but probing the heart of reionisation ($f_{\text{HI}} \sim 1$) directly is impossible using τ_{GP} alone.

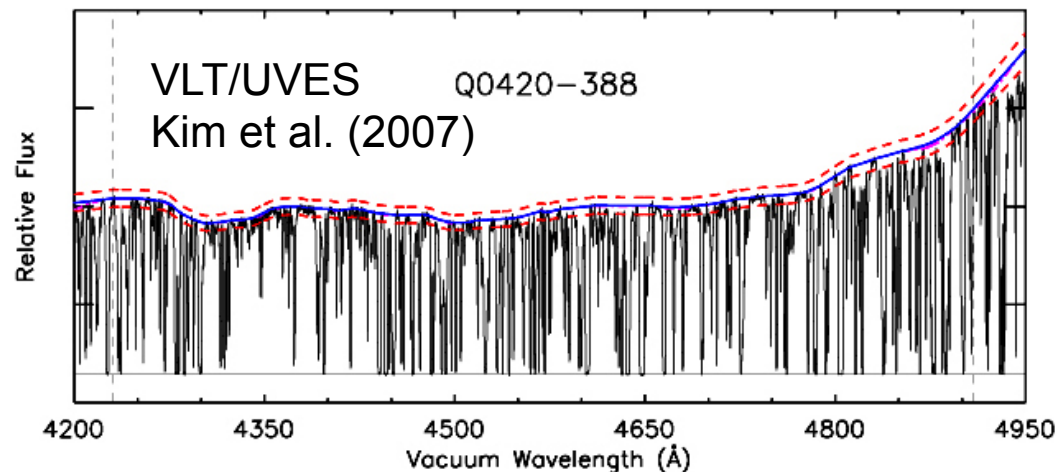
$$\tau_{\text{GP}} \approx 4.3 \times 10^5 \left(\frac{n_{\text{HI}}}{n_{\text{H}}} \right)$$



Fan et al. (2006)

Evidence from the Ly- α Forest

II. Measurements of the UV background at $z < 6$

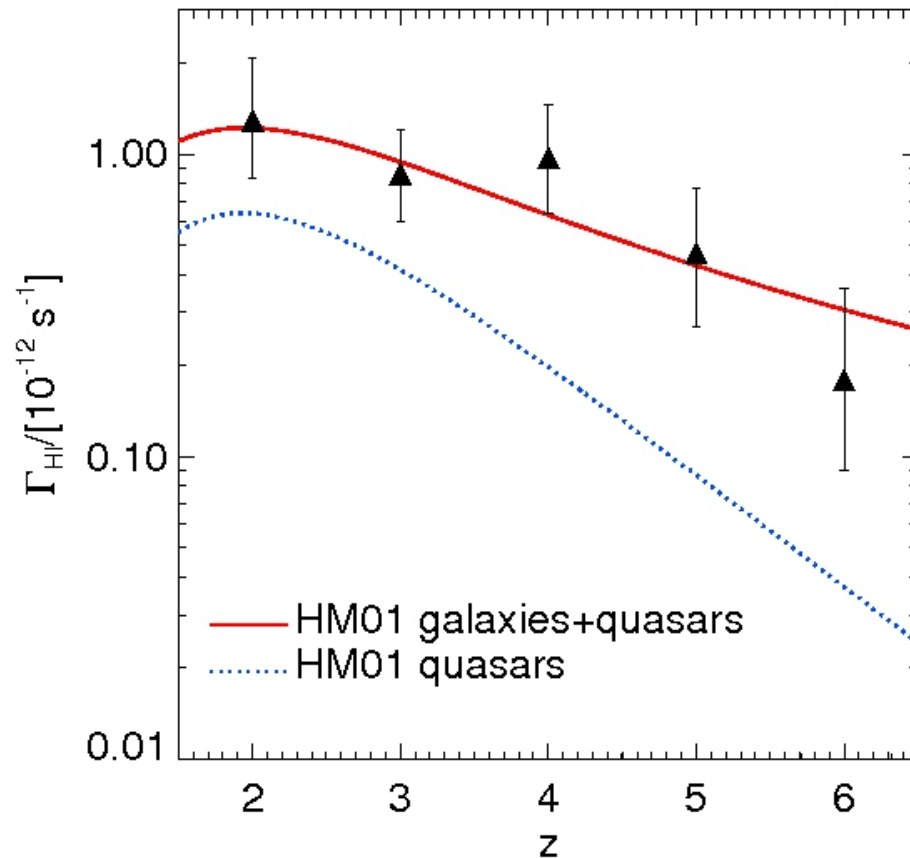


Constraints come from measurements of the opacity of the Ly- α forest and the line-of-sight proximity effect.

Ly- α opacity $\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T_0^{0.7} H(z) \Gamma(z)} (1+\delta)^{2-0.7(\gamma-1)}$

Photo-ionisation rate

The contribution of quasars



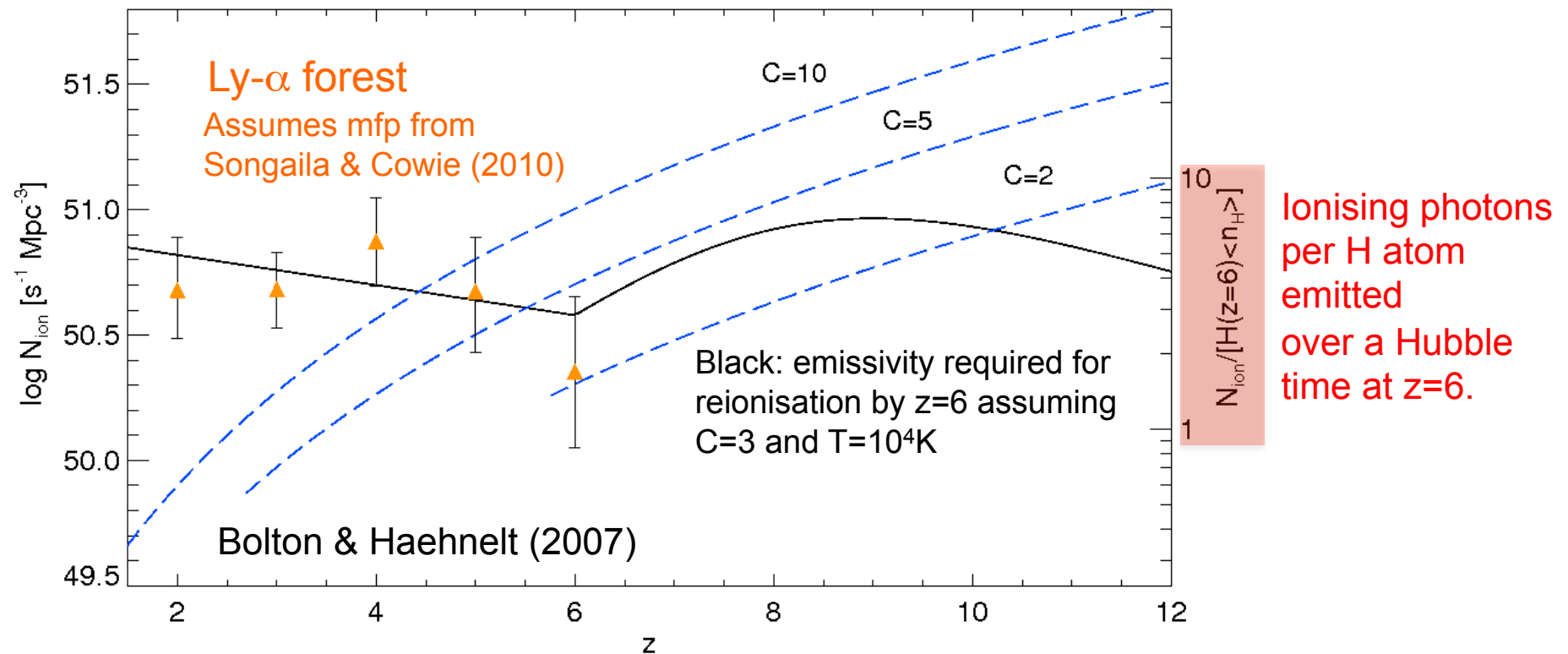
- Observed quasars at $z \sim 6$ do not account for all of the photons required to keep the Universe ionised at $z=6$.
- An increasingly significant contribution from star forming galaxies is required to maintain the UVB photon budget approaching $z=6$.

Bolton & Haehnelt (2007)

see also Faucher-Giguere et al (2008), Haardt & Madau (2011)

Evidence from the Ly- α Forest

III. A constraint on the required ionising emissivity at $z > 6$



- UV background consistent with only 1-3 ionising photons per HI emitted over a Hubble time at $z=6$ (see also Miralda-Escude 2003, Meiksin 2005).
- The paucity of ionising photons at $z=6$ means the emissivity has to remain constant or rise at $z > 6$ (depending on the ionising spectrum) for reionisation to complete by $z=6$.

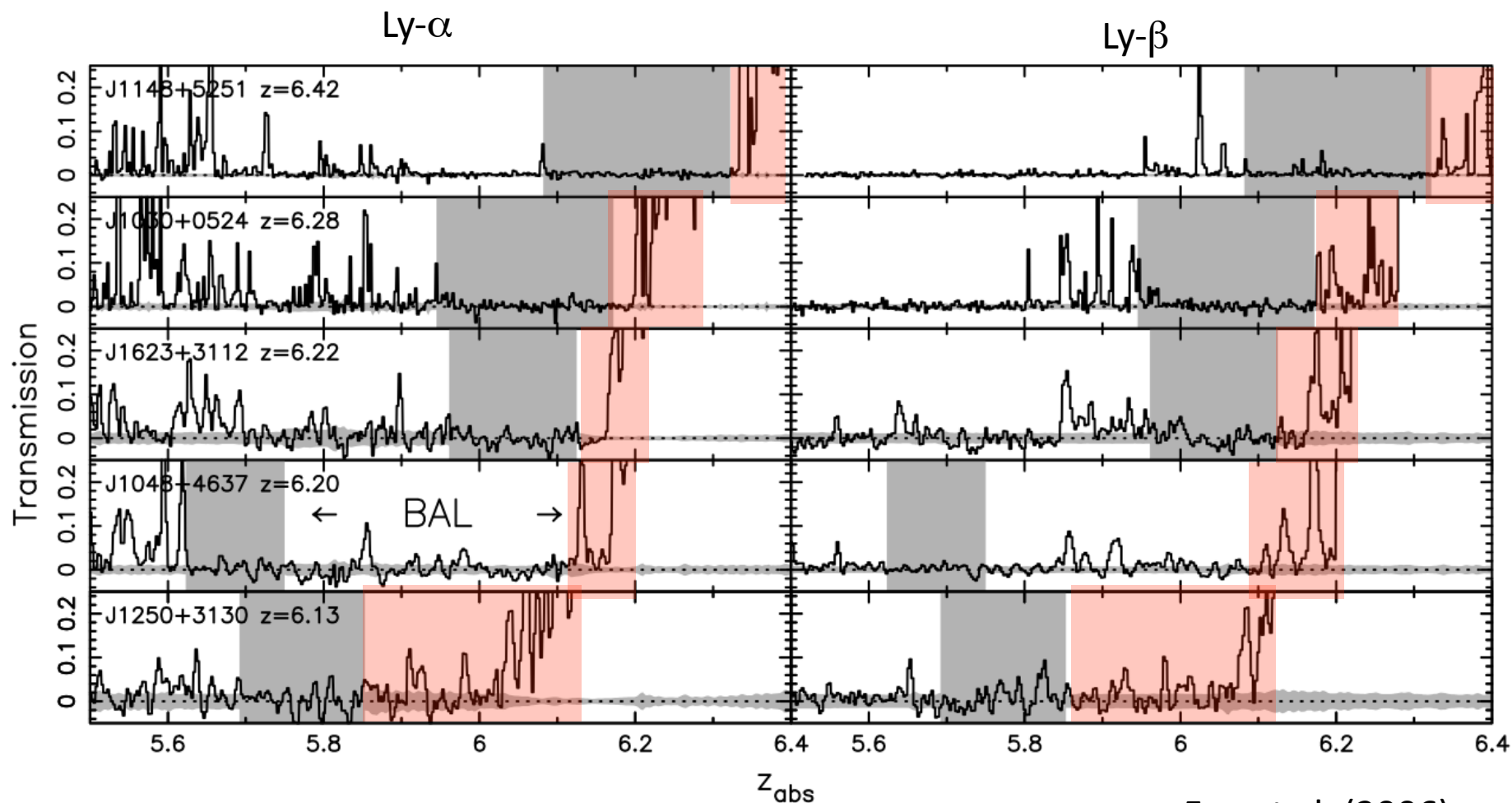
Alternative probes using Ly- α absorption

Can we learn any more about reionisation from Ly- α absorption?

- **Quasar near-zones** (Wyithe & Loeb 2004, Bolton & Haehnelt 2007, Lidz et al. 2007, Maselli et al 2007, 2009, Alavarez & Abel 2007, Mesinger & Haiman 2004, 2007, Wyithe et al. 2008, Mortlock et al. 2011)
- **Ly- α absorption line widths** (Theuns et al. 2002, Hui & Haiman 2003, Bolton et al. 2010)

Quasar near-zones

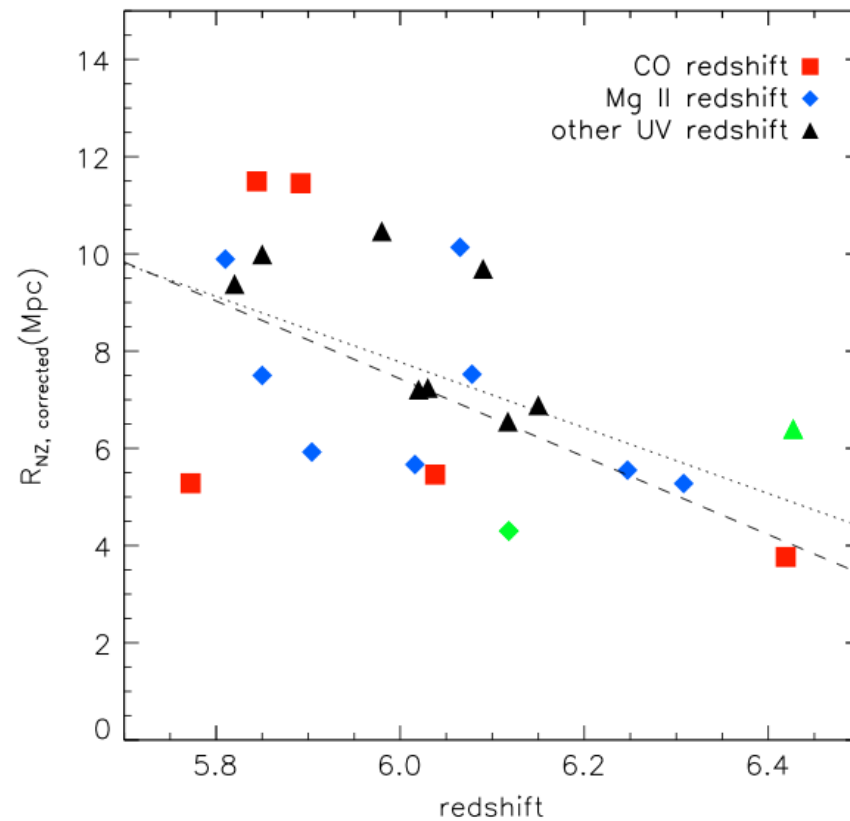
Near zones are the transmission windows blueward of Ly- α and redward of the Gunn-Peterson trough due to enhanced IGM ionisation in close proximity to the QSO.



Fan et al. (2006)

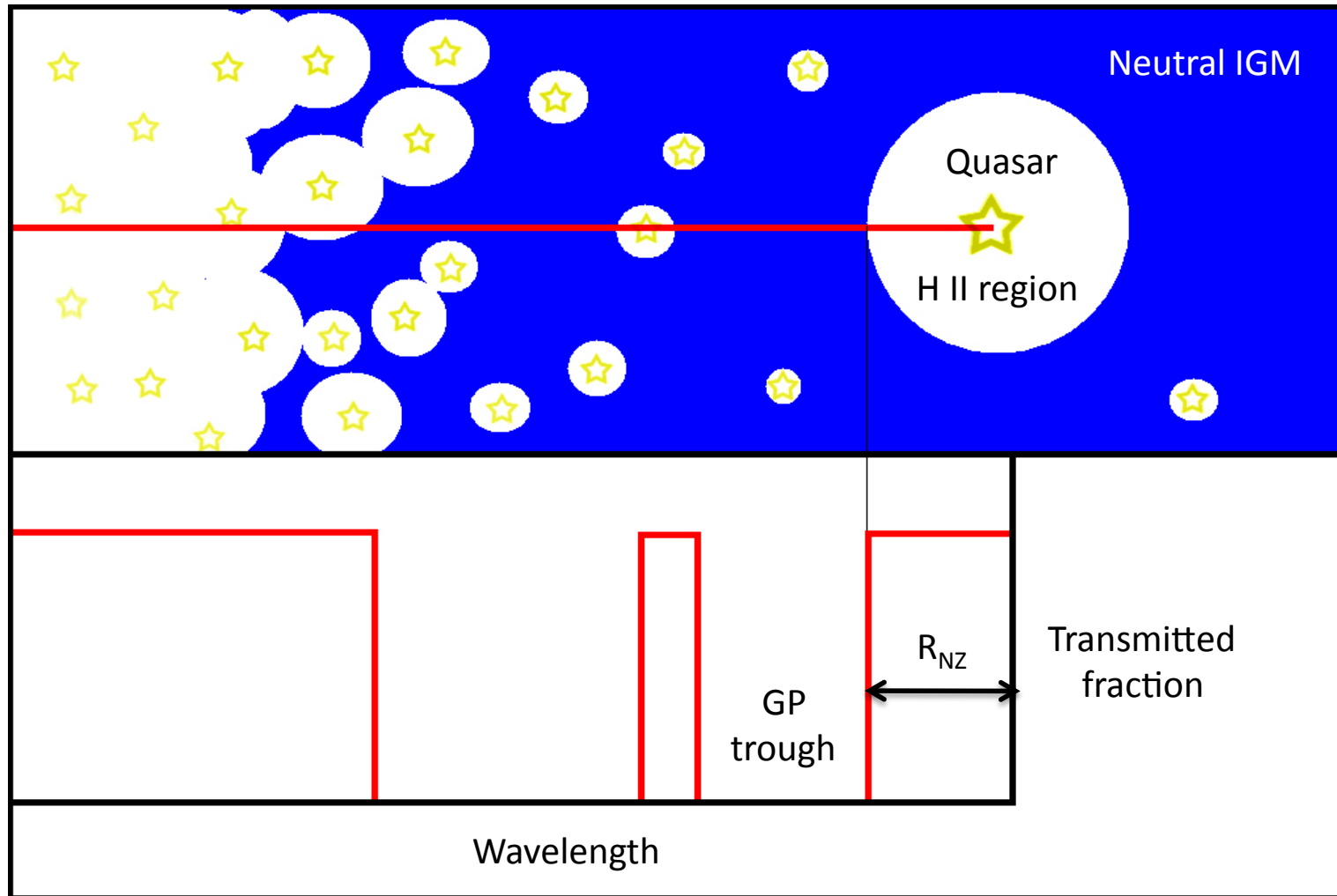
Near zone size with redshift

A trend of increasing near-zone size with decreasing redshift is observed at $5.7 < z < 6.4$. What drives this evolution?



Carilli et al. (2010)

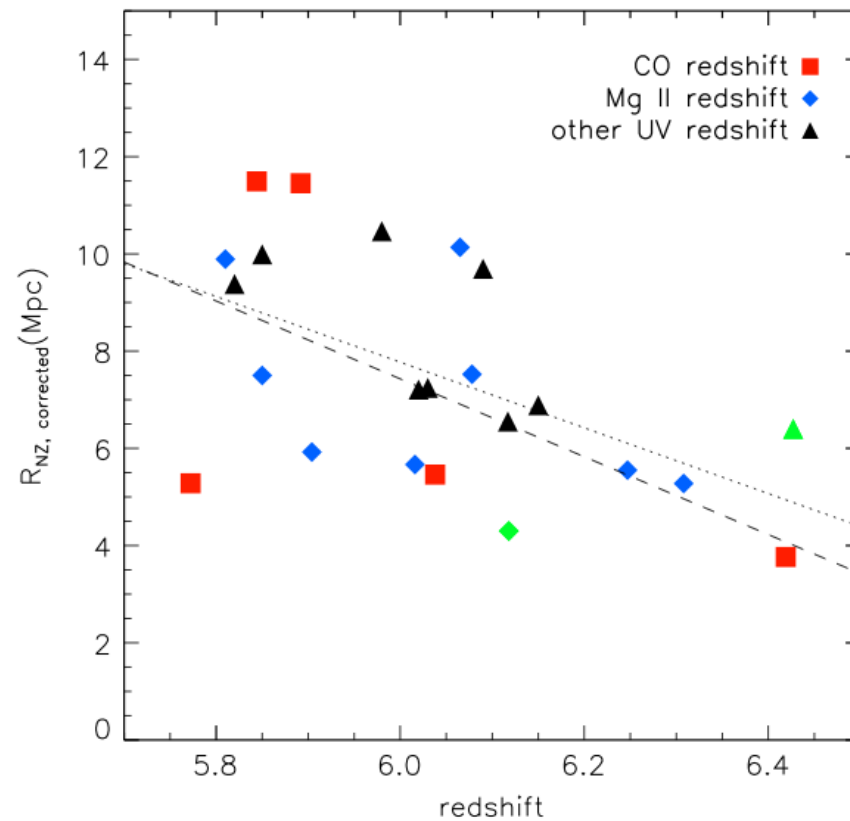
H-II regions...



$$R_{NZ} \propto \dot{N}^{1/3} t_Q^{1/3} f_{HI}^{-1/3} \quad \text{e.g. Wyithe \& Loeb (2004)}$$

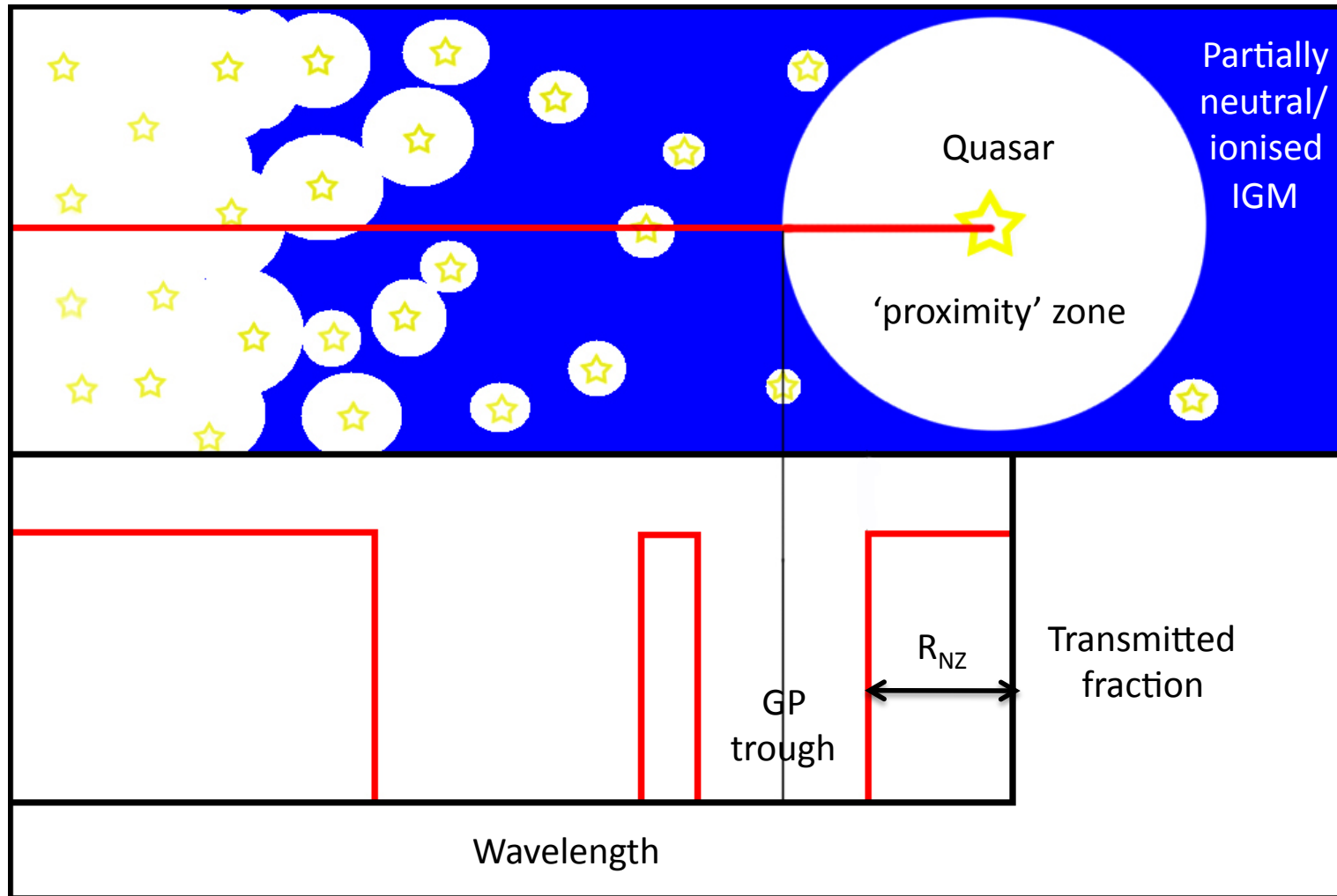
Near zone size with redshift

Assuming measured sizes correspond to the edges H-II regions, this would imply a factor of ~ 10 increase in f_{HI} from $z=5.8$ to $z=6.4$, but is this assumption reliable?



Carilli et al. (2010)

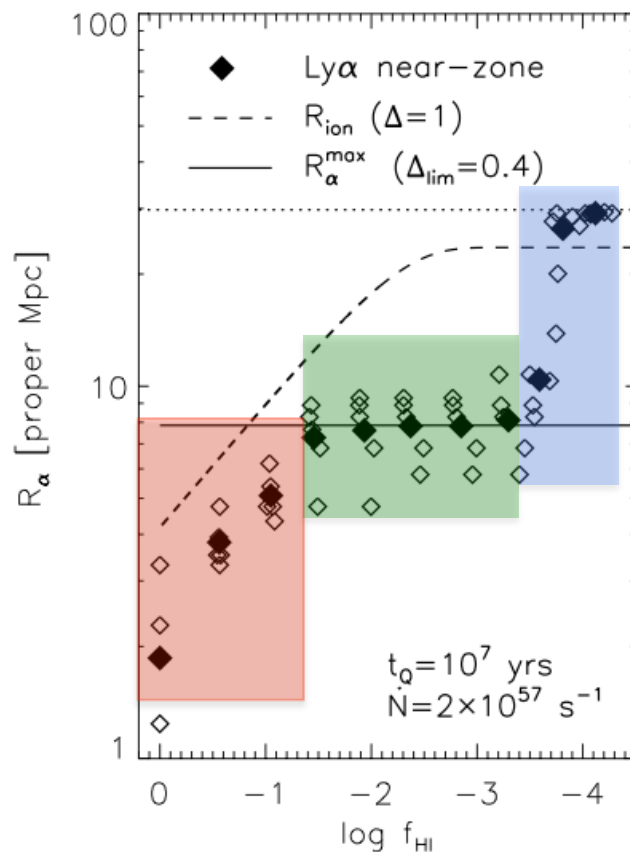
...or resonant absorption?



$$R_{NZ} \propto \dot{N}^{1/2}$$

Near-zone sizes from numerical simulations

Hydrodynamical simulations of structure formation combined with line-of-sight radiative transfer. The IGM neutral fraction near the quasar is varied, while the quasar age and ionising luminosity are held constant.



Bolton & Haehnelt (2007)

1) Size corresponds to H-II region extent

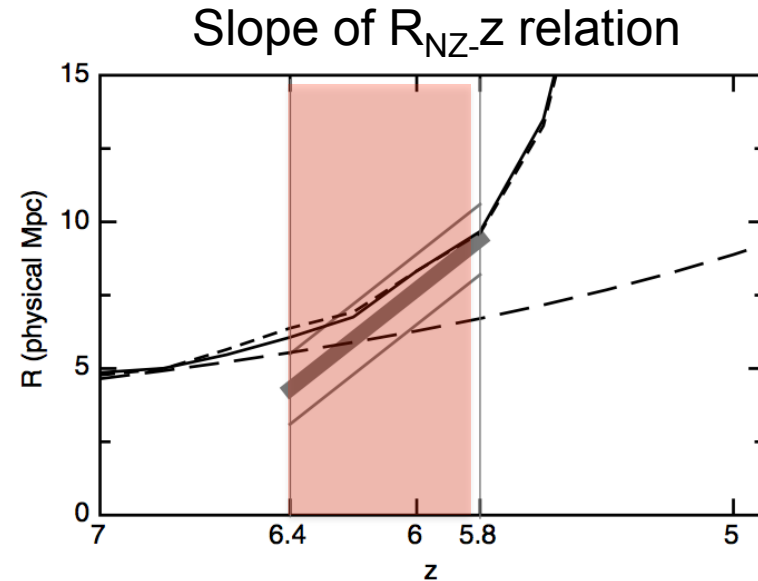
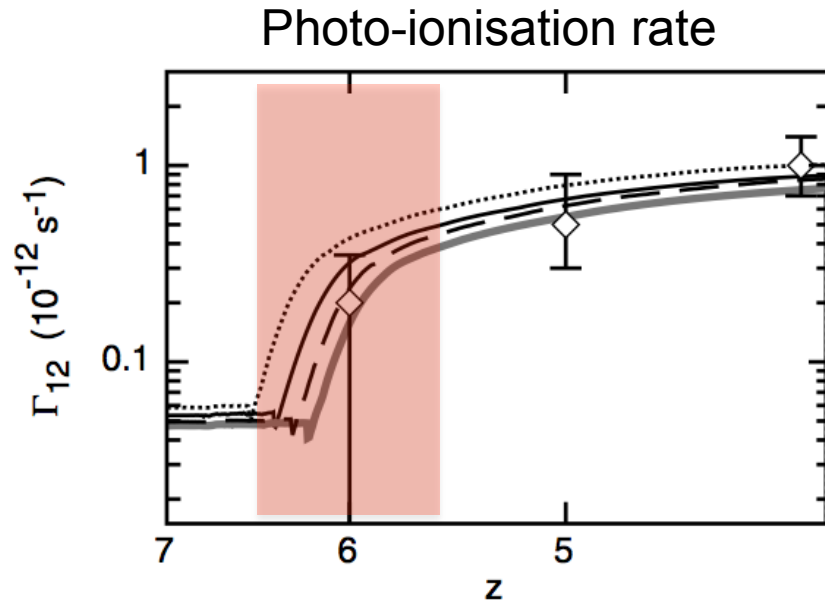
$$R_{NZ} \propto \dot{N}^{1/3} t_Q^{1/3} f_{HI}^{-1/3}$$

2) Size set by resonant absorption

$$R_{NZ} \propto \dot{N}^{1/2}$$

3) Size set by transmission from thinning Ly- α forest due to the rising background photo-ionisation rate (i.e. the disappearing GP trough).

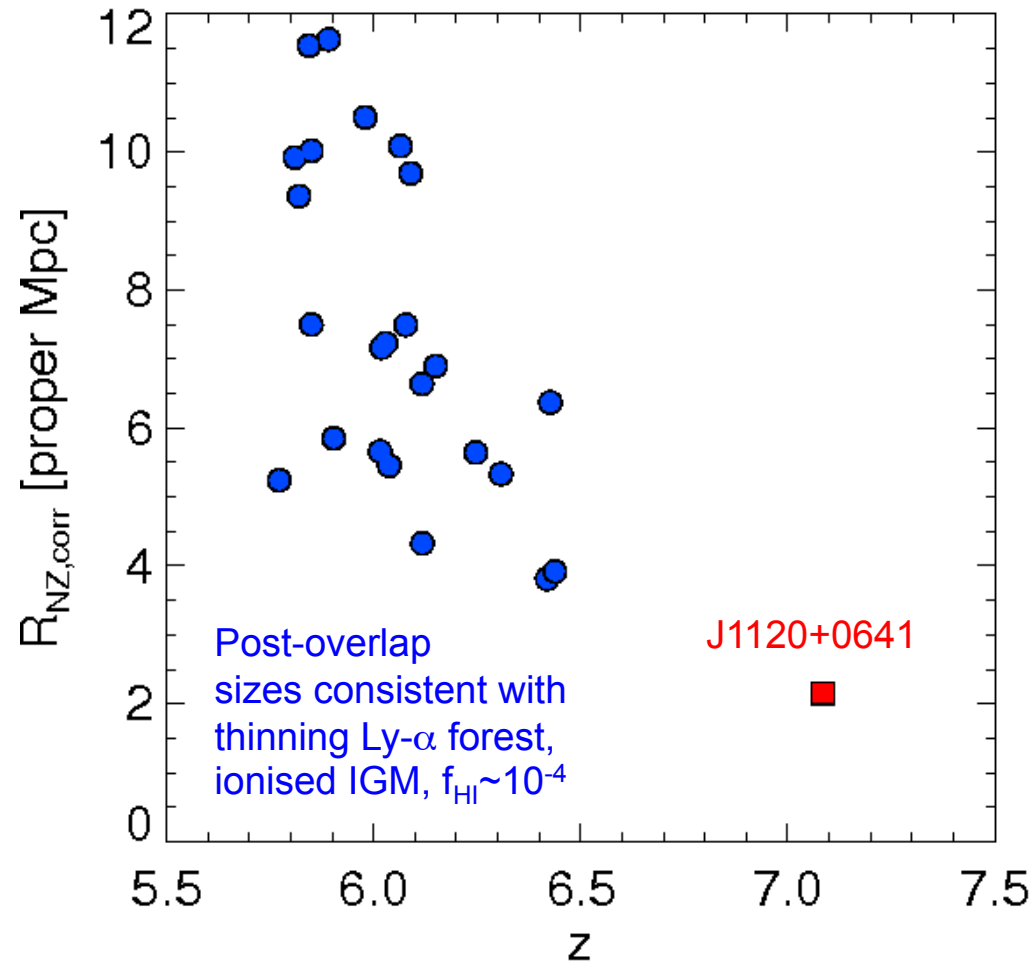
Explaining the size-redshift relation at $5.7 < z < 6.4$



Wyithe et al. (2008)

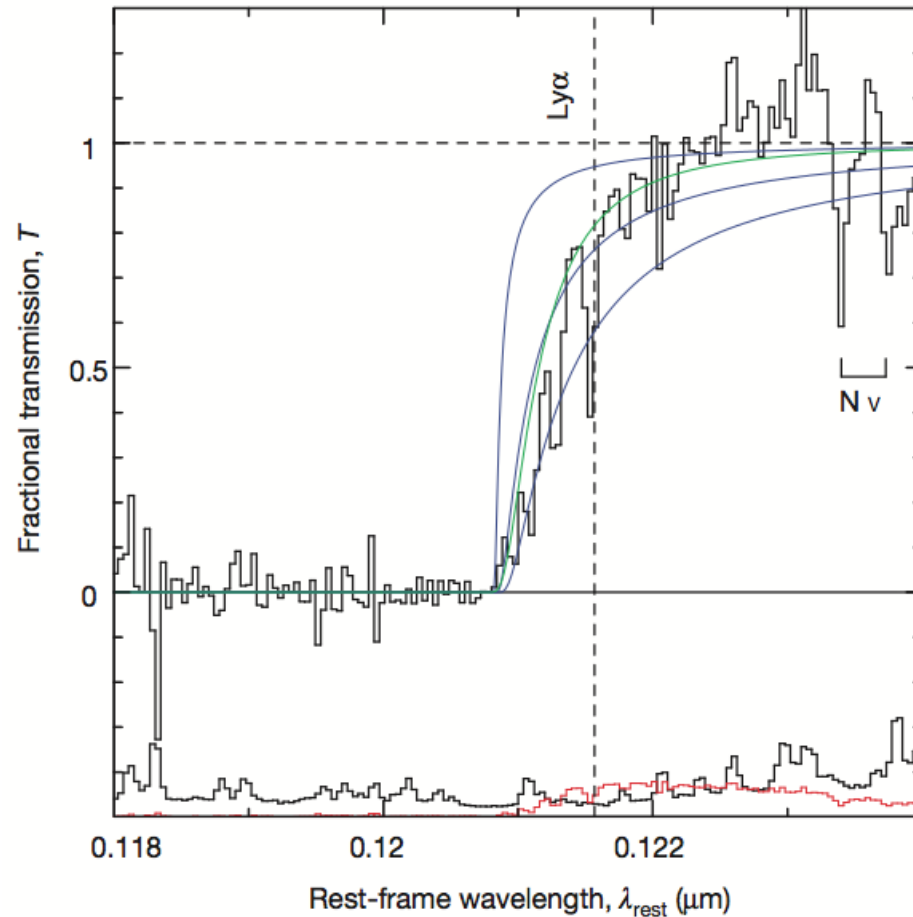
- Redshift evolution at $5.7 < z < 6.4$ is driven by the **increasing amplitude of the ionising background**, due to the rapidly rising ionising photon mean free path at the end of reionisation following H-II bubble “overlap”.
- The $z \sim 6$ near-zone sizes and their redshift dependence are consistent with a highly ionised IGM, $f_{\text{HI}} \sim 10^{-3} - 10^{-4}$ (see also Maselli et al. 2009).

ULAS J1120+0641 at $z=7.085$



ULAS J1120+0641 exhibits a rather **small near-zone** (1.9 proper Mpc) for its bright magnitude ($M_{1450} = -26.6$)

ULAS J1120+0641 at $z=7.085$

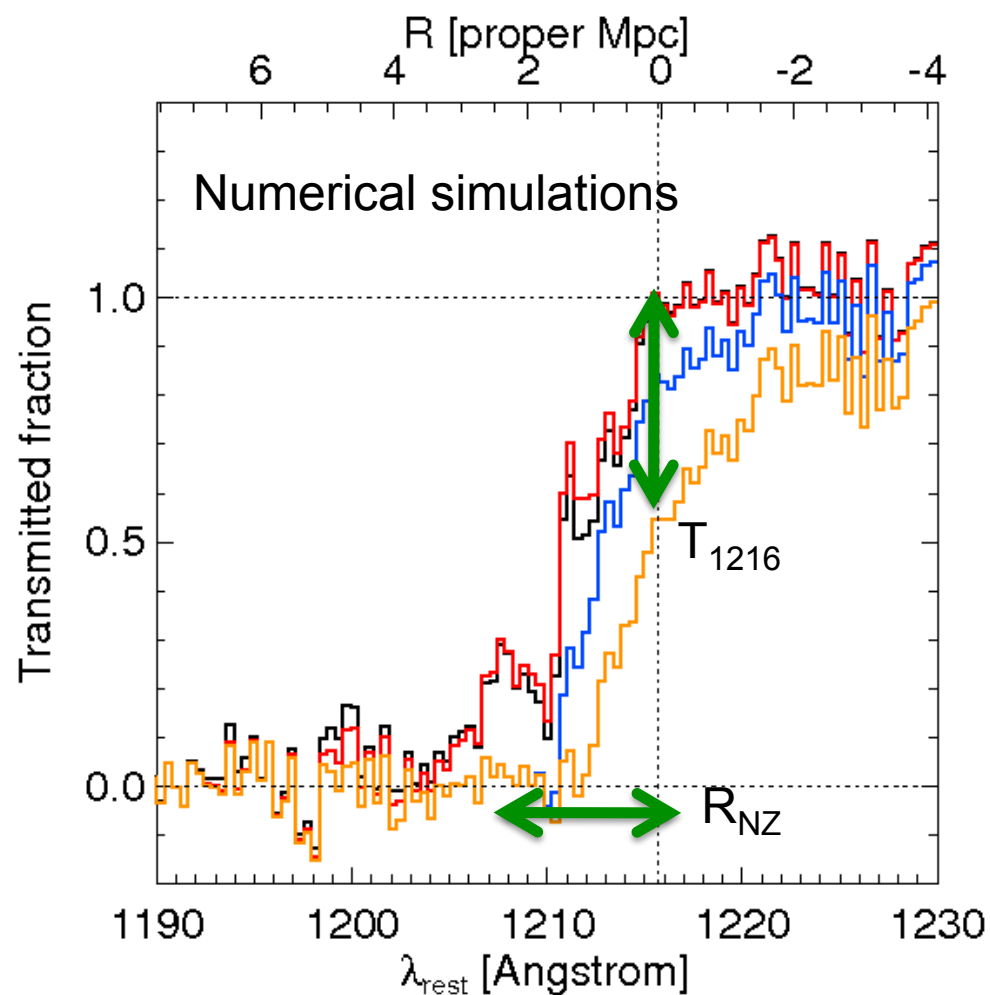


- The spectrum also exhibits a smooth absorption component redward of the $\text{Ly}-\alpha$ emission line, consistent with an IGM damping wing.

(e.g. Miralda-Escude 1998, Mesinger & Haiman 2004)

Mortlock et al. (2011)

How neutral is the IGM around J1120+0641?



Bolton et al. (2011)

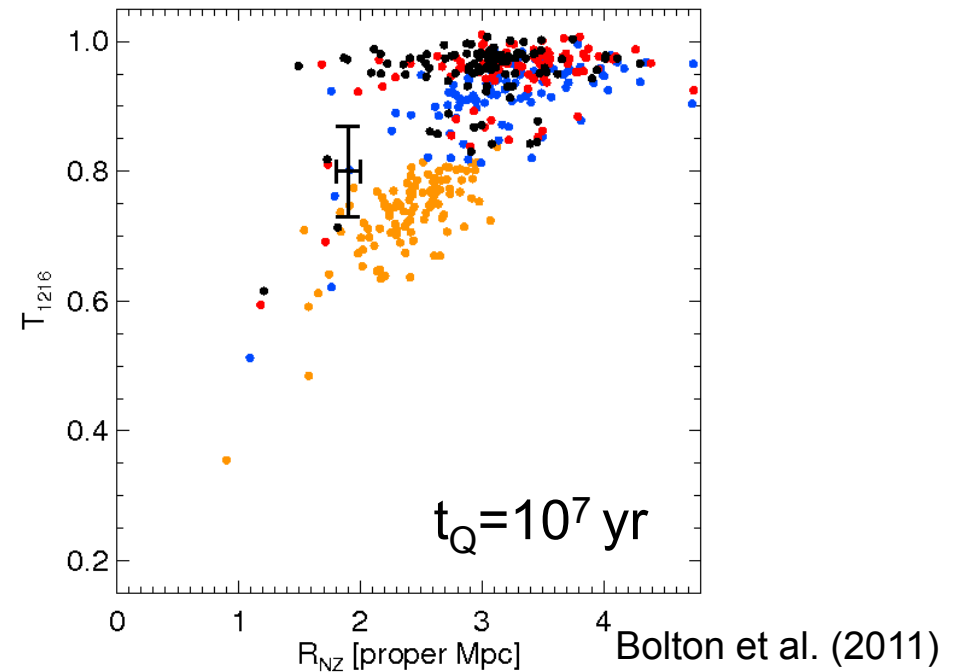
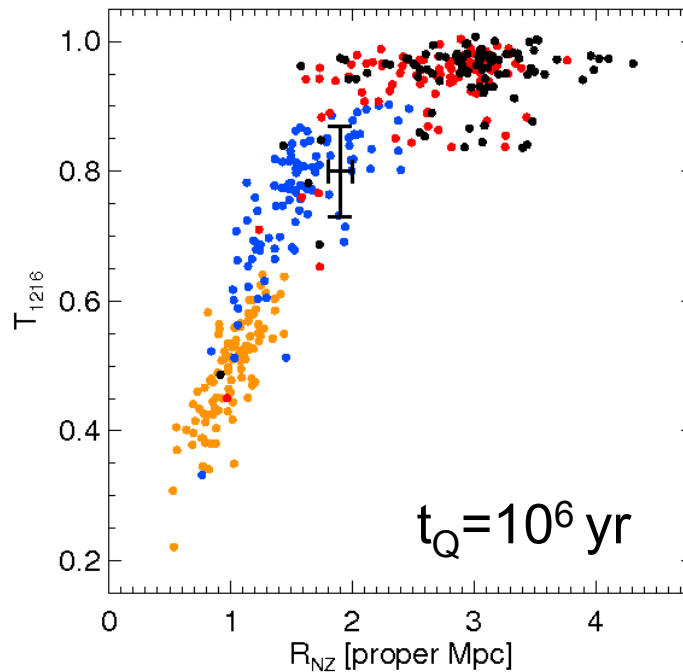
- Hydrodynamical simulations of the inhomogeneous IGM + line-of-sight Lyman continuum radiative transfer.

- Consider two quantities which characterise the near-zone, R_{NZ} and T_{1216} , and vary f_{HI} and duration of optically bright phase, t_Q .

$$f_{\text{HI}} \approx \begin{array}{l} 5.0 \times 10^{-4} \\ 6.2 \times 10^{-3} \\ 1.1 \times 10^{-1} \\ 1.0 \end{array}$$

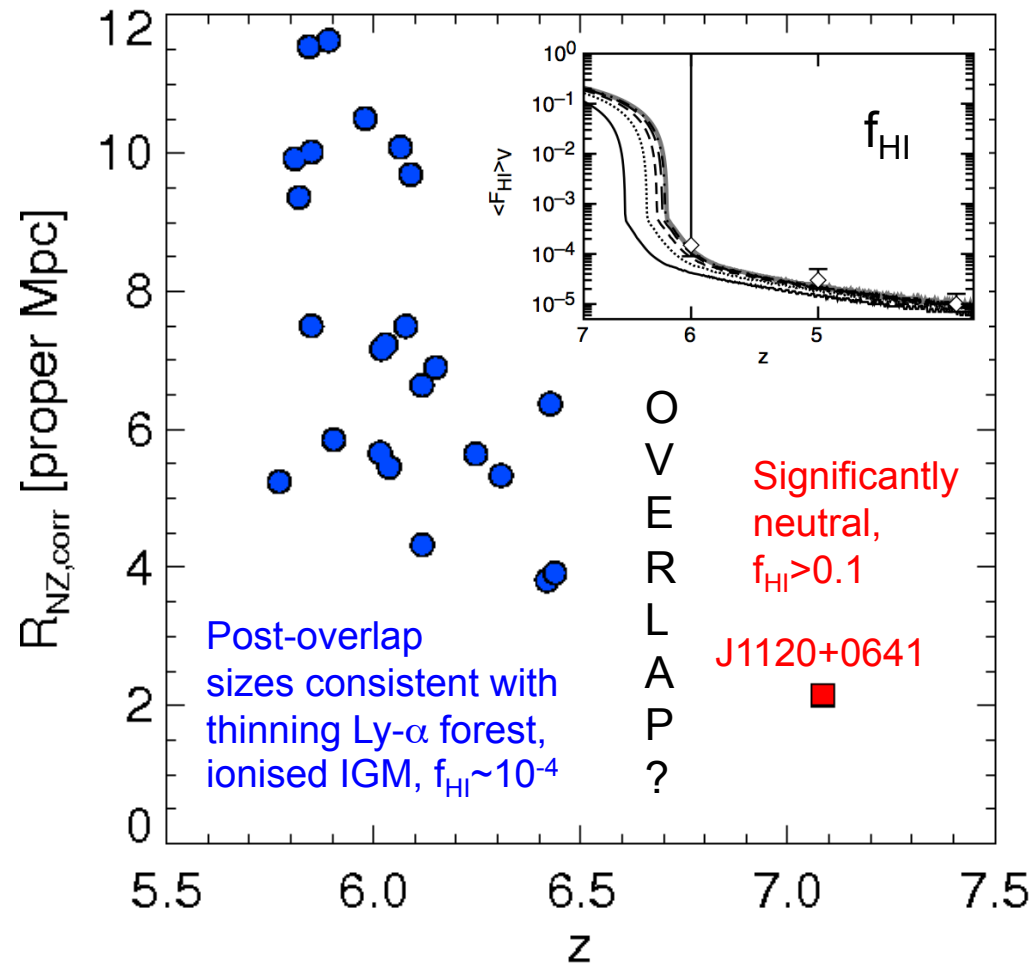
How neutral is the IGM around J1120+0641?

Transmitted fraction at rest frame Ly- α against near-zone size



- For an optically bright phase of 10^6 yr (10^7 yr) the near-zone size is consistent with $f_{\text{HI}} \sim 0.1$ (~ 1.0)
- A highly ionised IGM is also consistent with around 5% of simulated sight-lines which exhibit a proximate DLA.
- Such a short optically bright phase suggests that the quasar may grow predominantly in an optically obscured phase.

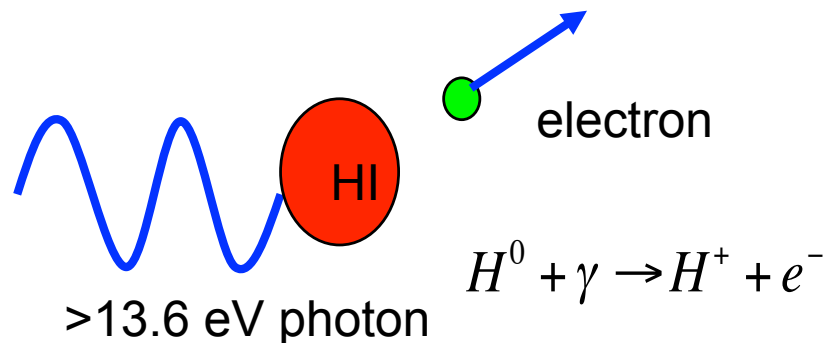
Implications for the reionisation history?



Caution: Reionisation is *inhomogeneous*, one object is not the full story (see e.g. Mesinger & Furlanetto 2008).

Reionisation and the thermal history

Reionisation photo-heats
the IGM



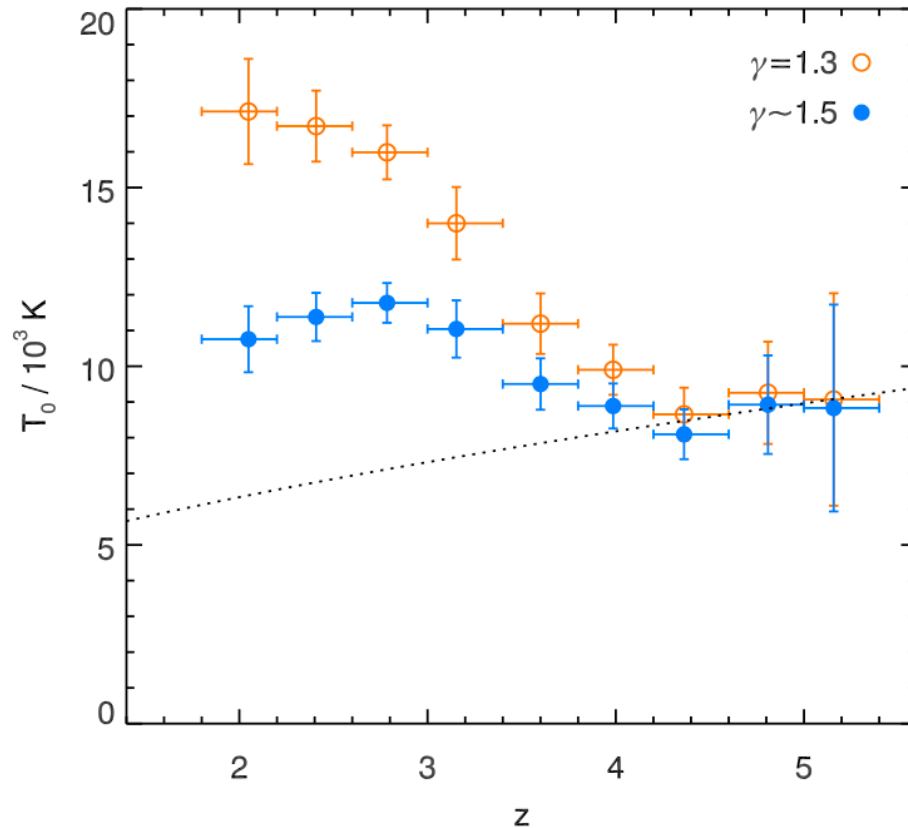
Long adiabatic cooling timescale enables use of temperature as an indirect probe of the H I and He II reionisation epochs.

The temperature of the IGM depends on:

- 1) When the IGM was reionised (how much time available to cool?)
- 2) The spectra of the ionising sources responsible for reionisation (harder spectra = more heating).

see e.g. Hui & Haiman (2003),
Theuns et al. (2002)

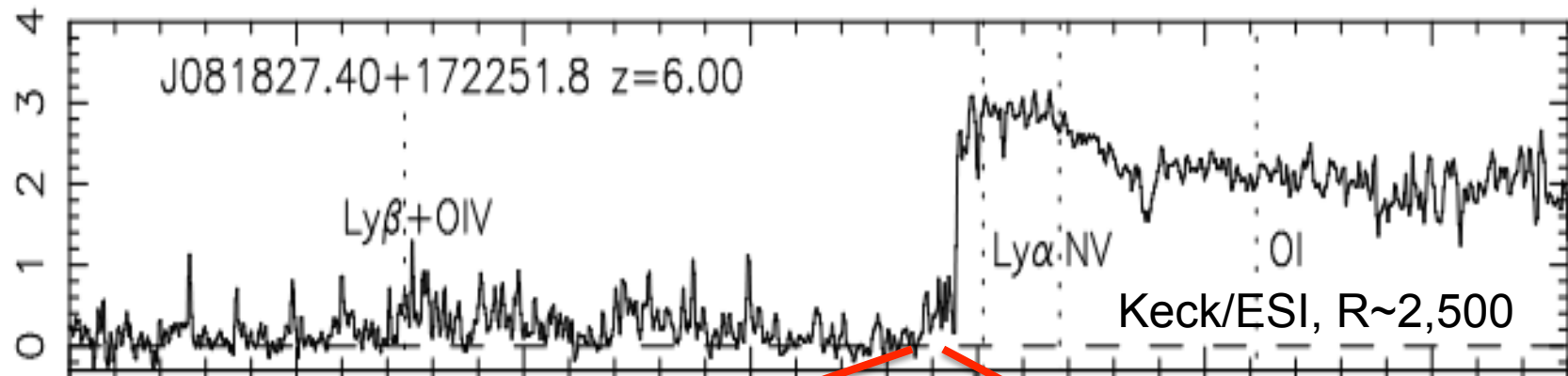
IGM temperature at $z < 5$



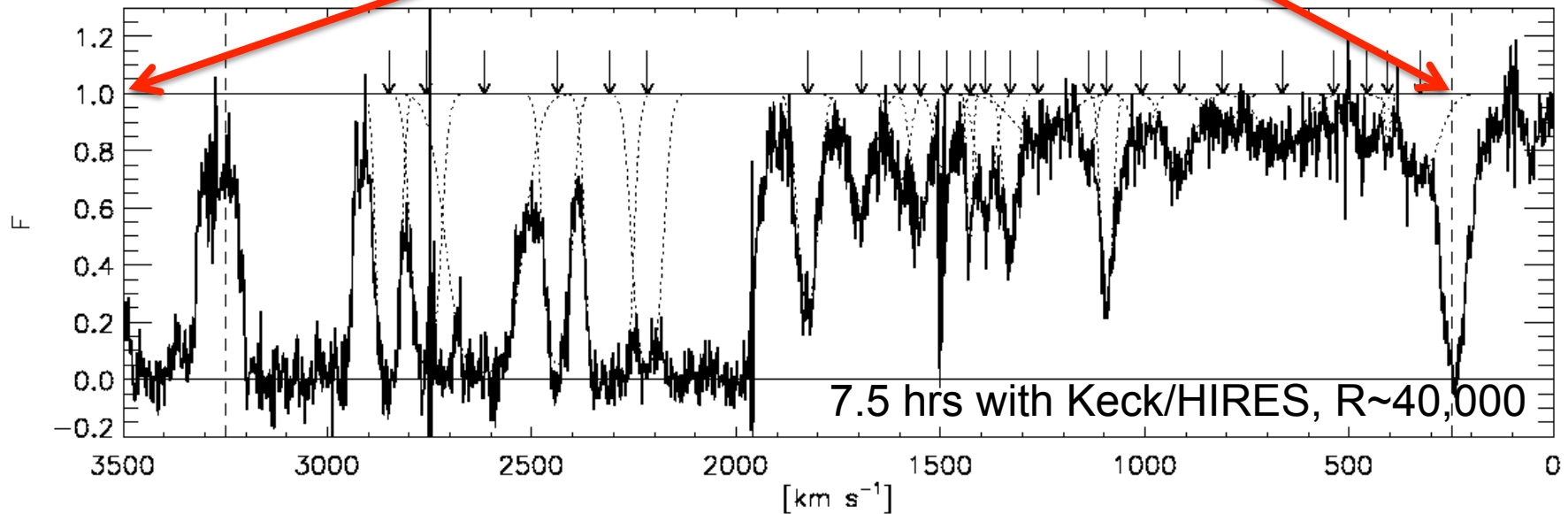
Becker et al. (2011)

- A clear rise in temperature towards $z=2$ from thermal broadening of Ly- α forest lines, signature of He-II reionisation at $z < 4.5$ (see also e.g. Schaye et al. 2000, Lidz et al. 2010).
- However, He-II heating complicates interpretation with respect to H I reionisation.
- Higher redshift, $z > 5$ constraints needed, but complicated by the rapidly increasing Ly- α opacity.

IGM temperature at $z \sim 6$

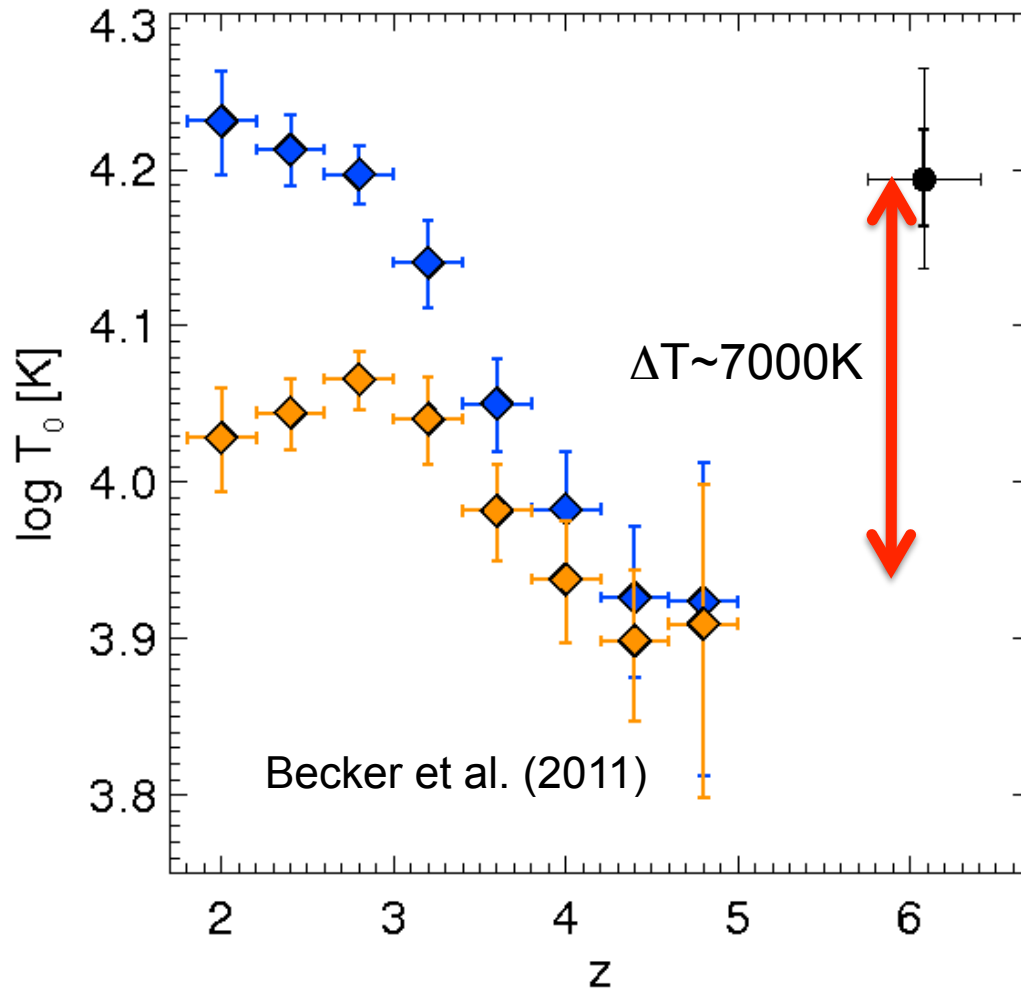


Fan et al. (2006)



Becker et al. (2007), Bolton et al. (2010)

He-II photo-heating at $z \sim 6$



Bolton et al. (in prep.)

- New near-zone temperature measurements from 7 Keck/HIRES spectra (Becker et al. 2007), significantly expanding the study by Bolton et al. (2010).
- The average temperature at mean density is $\sim 8000\text{K}$ hotter than the general IGM at $z \sim 5$.
- This additional heating is consistent with quasars with an EUV spectral index $\alpha=1.5$ locally reionising the He-II.

Summary

- Ly- α absorption in the spectra of high redshift quasars provides a wealth of information on the epoch of reionisation, in addition to the well known Gunn-Peterson trough constraints.
- The rapid evolution of Ly- α near-zone sizes with redshift at $5.7 < z < 6.4$ is driven by the rising intensity of the UV background at $z \sim 6$, expected towards the end of reionisation following overlap. This data are consistent with an IGM which still highly ionised at $z \sim 6$.
- The recently discovered quasar ULAS J1120+0641 at $z=7.085$ exhibits a damping wing and near zone size consistent with IGM neutral fraction $>10\%$, using the same models which successfully match the $z \sim 6$ near-zone data. This suggests the mean free path for ionising photons falls rapidly toward $z=7$, and that the final stages of reionisation are occurring at $z=6.5-7$.
- IGM temperature measurements in the near-zones of quasars at $z=6$ exhibit temperatures $\sim 8000\text{K}$ higher than the general IGM at $z \sim 5$, implying these luminous quasars are reionising He-II in their vicinity